

Regularization in Machine Learning

Introduction

- While developing machine learning models you must have encountered a situation in which the training accuracy of the model is high but the validation accuracy or the testing accuracy is low. This is the case which is popularly known as overfitting in the domain of machine learning.

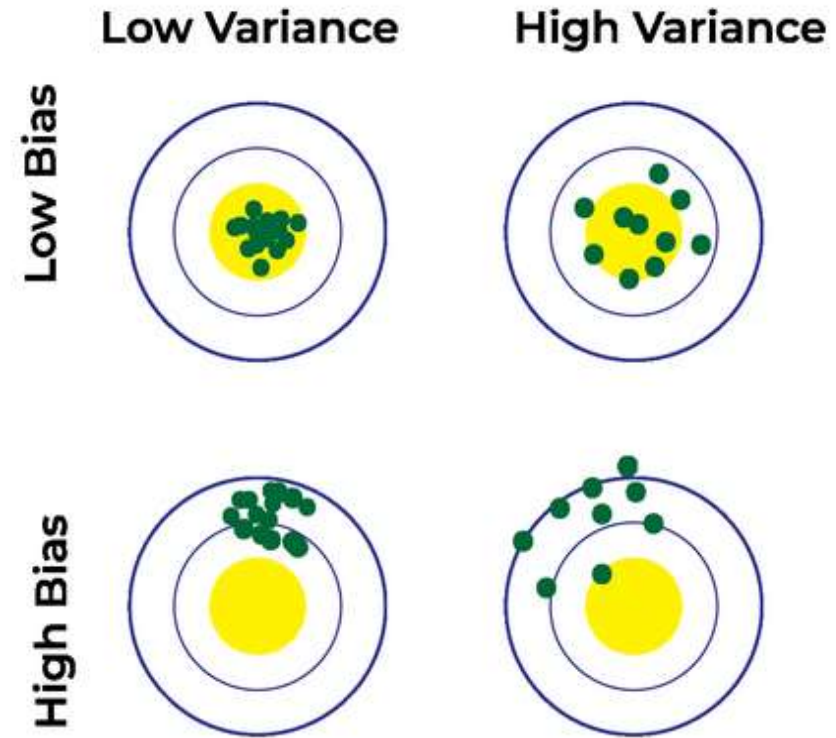
Role Of Regularization

- Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function, discouraging the model from assigning too much importance to individual features or coefficients.
- **Complexity Control:** Regularization helps control model complexity by preventing overfitting to training data, resulting in better generalization to new data.
- **Preventing Overfitting:** One way to prevent overfitting is to use regularization, which penalizes large coefficients and constrains their magnitudes, thereby preventing a model from becoming overly complex and memorizing the training data instead of learning its underlying patterns.

- **Balancing Bias and Variance:** Regularization can help balance the trade-off between model bias (underfitting) and model variance (overfitting) in machine learning, which leads to improved performance.
- **Feature Selection:** Some regularization methods, such as L1 regularization (Lasso), promote sparse solutions that drive some feature coefficients to zero. This automatically selects important features while excluding less important ones.

- **Handling Multicollinearity:** When features are highly correlated (multicollinearity), regularization can stabilize the model by reducing coefficient sensitivity to small data changes.
- **Generalization:** Regularized models learn underlying patterns of data for better generalization to new data, instead of memorizing specific examples.

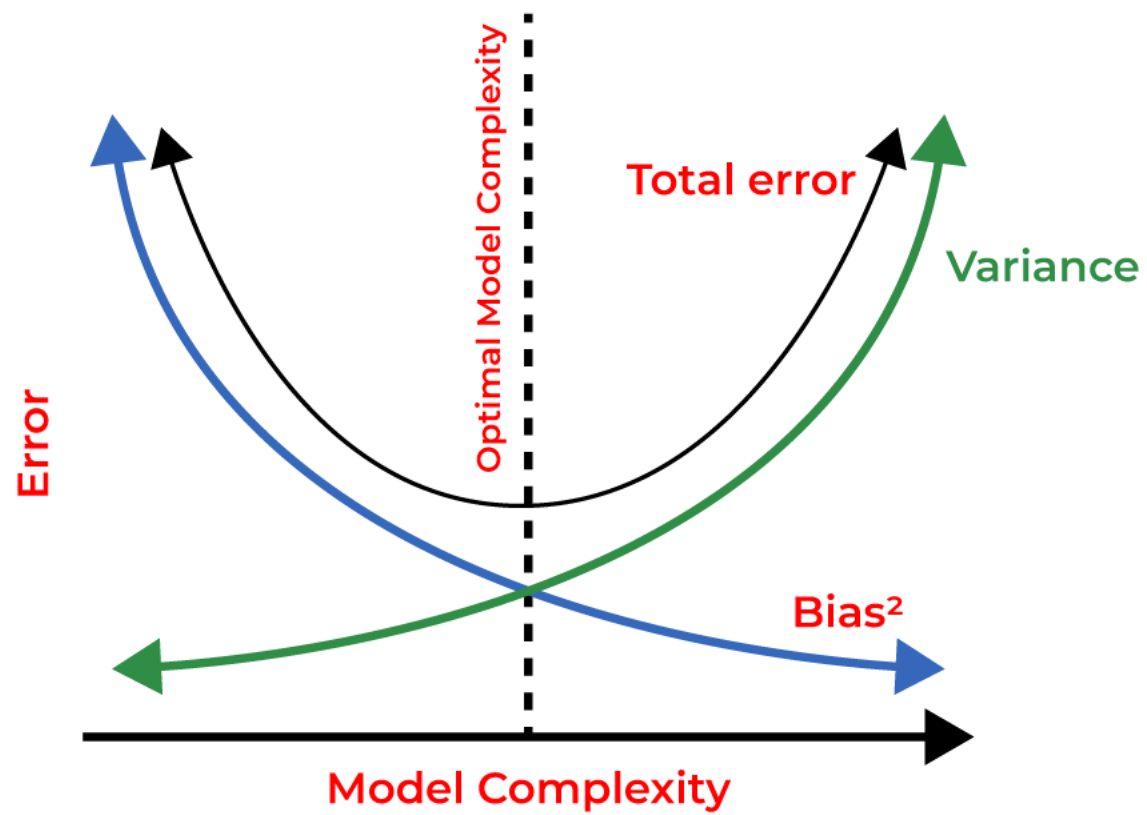
Different Combinations of Bias-Variance



- **High Bias, Low Variance:** A model that has high bias and low variance is considered to be underfitting.
- **High Variance, Low Bias:** A model that has high variance and low bias is considered to be overfitting.
- **High-Bias, High-Variance:** A model with high bias and high variance cannot capture underlying patterns and is too sensitive to training data changes. On average, the model will generate unreliable and inconsistent predictions.
- **Low Bias, Low Variance:** A model with low bias and low variance can capture data patterns and handle variations in training data. This is the perfect scenario for a machine learning model where it can generalize well to unseen data and make consistent, accurate predictions. However, in reality, this is not feasible.

Bias Variance tradeoff

- The bias-variance tradeoff is a fundamental concept in machine learning. It refers to the balance between bias and variance, which affect predictive model performance. Finding the right tradeoff is crucial for creating models that generalize well to new data.
- The bias-variance tradeoff demonstrates the inverse relationship between bias and variance. When one decreases, the other tends to increase, and vice versa.
- Finding the right balance is crucial. An overly simple model with high bias won't capture the underlying patterns, while an overly complex model with high variance will fit the noise in the data.



Regularization in Machine Learning

- Regularization is a technique used to reduce errors by fitting the function appropriately on the given training set and avoiding overfitting. The commonly used regularization techniques are :
 - Lasso Regularization – L1 Regularization
 - Ridge Regularization – L2 Regularization
 - Elastic Net Regularization – L1 and L2 Regularization

Lasso Regression

A regression model which uses the **L1 Regularization** technique is called **LASSO(Least Absolute Shrinkage and Selection Operator)** regression. **Lasso Regression** adds the “*absolute value of magnitude*” of the coefficient as a penalty term to the loss function(L). Lasso regression also helps us achieve feature selection by penalizing the weights to approximately equal to zero if that feature does not serve any purpose in the model.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^m |w_i|$$

where,

- ***m*** – Number of Features
- ***n*** – Number of Examples
- ***y_i*** – Actual Target Value
- ***y_i(hat)*** – Predicted Target Value

Ridge Regression

A regression model that uses the **L2 regularization** technique is called **Ridge regression**. **Ridge regression** adds the “*squared magnitude*” of the coefficient as a penalty term to the loss function(L).

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^m w_i^2$$

Elastic Net Regression

This model is a combination of L1 as well as L2 regularization. That implies that we add the absolute norm of the weights as well as the squared measure of the weights. With the help of an extra hyperparameter that controls the ratio of the L1 and L2 regularization.

$$\text{Cost} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \left((1 - \alpha) \sum_{i=1}^m |w_i| + \alpha \sum_{i=1}^m w_i^2 \right)$$

Benefits of Regularization

- Regularization improves model generalization by reducing overfitting. Regularized models learn underlying patterns, while overfit models memorize noise in training data.
- Regularization techniques such as L1 (Lasso) L1 regularization simplifies models and improves interpretability by reducing coefficients of less important features to zero.
- Regularization improves model performance by preventing excessive weighting of outliers or irrelevant features.
- Regularization makes models stable across different subsets of the data. It reduces the sensitivity of model outputs to minor changes in the training set.

- Regularization prevents models from becoming overly complex, which is especially important when dealing with limited data or noisy environments.
- Regularization can help handle multicollinearity (high correlation between features) by reducing the magnitudes of correlated coefficients.
- Regularization introduces hyperparameters (e.g., α or λ) that control the strength of regularization. This allows fine-tuning models to achieve the right balance between bias and variance.
- Regularization promotes consistent model performance across different datasets. It reduces the risk of dramatic performance changes when encountering new data.